

Music Genre Classification

Bachelor Semester Project S2 (Academic Year 2024/25), University of Luxembourg

Ene Alexandra
alexandra.ene.001@student.uni.lu
Université du Luxembourg

Luis A. Leiva
luis.leiva@uni.lu
Université du Luxembourg

ABSTRACT

Automatic music genre classification helps organize and find songs more easily in large digital music libraries. Prior efforts have explored audio feature-based classification using various machine learning algorithms, but the impact of class imbalance and dataset size still poses problems in classification. Here we show that advanced classifiers such as support vector machines and multi-layer perceptrons outperform simpler models in most cases, particularly when scaling from 15 to 151 music genres. These findings suggest that non-linear, high-capacity models are better suited for genre classification tasks involving noisy, overlapping data, created by smaller classes with similar features that might be mistaken for other bigger classes. Our results help show that class balance and the way features are used can make a difference when trying to classify songs into many genres.

1 INTRODUCTION

A music genre is a conventional category a song might belong to, based on various factors such as BPM, liveliness, valence, etc.

Music genre classification has become a recurring subject in the modern day. It is one of the branches of Music Information Retrieval (MIR)[1], offering both theoretical and practical benefits. As musical libraries continue to expand, effective methods that need to be used for correctly classification, become increasingly important.

From a commercial perspective, genre classification enhances the performance of recommendation systems used by streaming platforms such as Spotify, YouTube Music or Apple Music. By identifying the genre of a song, these apps can offer an enhanced user experience.

Genre classification is also used in other fields such as sociological studies of music[2][3]. It helps us have a better view of the historical evolution of music styles, the influence of social movements, and the formation of musical identities. Moreover, its applications extend into health-related domains, including music therapy[4]. In medical environments music is often selected based on its emotional, cultural, and psychological impact on the individual. Since musical preferences are closely tied to genre, the ability to classify and retrieve music by genre becomes a valuable tool in therapeutic interventions.

Our study focuses on evaluating and comparing various classification models by analyzing their performance using multiple accuracy metrics and evaluation scores. The objective is to compare them in order to find in what context some of the classifiers outperform others. Furthermore, by conducting experiments on both a smaller and a larger dataset, we identify patterns and variations in the result changes.

2 RELATED WORK

Music genre classification has gained more and more popularity in recent years. Numerous studies have gone in depth into this topic, and began to further research it.

One of the studies that focuses on the same dataset as the one we are looking at, is titled "Hit or miss? Predicting Spotify track success from audio features[5]." To analyze the data, the author uses the median value of the "ms_played" variable as a threshold: tracks exceeding this value were labeled as "hits," while those below were categorized as "flops." This binary classification enabled the application of a supervised machine learning model aimed at distinguishing between successful and unsuccessful songs based solely on their audio features. The resulting model achieved a classification accuracy of 57%, with a precision of 56% and a recall of 60% for the "hit" category. Analysis of the confusion matrix revealed notable misclassifications. Looking at this study, we can see decent results in the scores, while some classifiers from our study present lower results, others show a much better performance.

"EDA Artists and Genres [6]" that focuses on exploratory data analysis of a music dataset, examining relationships between artists and genres. The primary objective is to uncover patterns and insights that could inform further analyses or applications in music. While not directly classifying the data with the help of machine learning algorithms, it gives us a deeper insight into the song distribution.

This may be further researched in the study "Spotify Song Attributes Analysis[7]" which focuses even more on the relationship and correlation between attributes.

A study that focuses on a similar dataset is "Spotify Tracks Dataset [8]", that explores the use of machine learning to predict song popularity and build a music recommendation system. Among the models tested, logistic regression yielded poor results, while k-nearest neighbors improved performance, achieving around 70% accuracy and very high recall for identifying top songs. However, the best results came from a decision tree classifier, which reached an accuracy of about 72.3%, with 99% recall and 90% F1-score for predicting highly popular songs. The decision tree also had a strong AUC score of 0.95, showing its ability to effectively separate the different popularity classes.

Another study worth mentioning is "Predicting the Popularity of Spotify Songs Using a Neural Network[9]" that focuses on how well a neural network can predict whether a Spotify song will be popular or not, compared to other machine learning models like random forest and support vector machine. This study, even though it's not made on the same dataset, it gives us an insight into the efficiency of the SVM and random forest, two classifiers that were also used in our study, and shows them consistently working and giving high results.

The article “Survey on deep learning with class imbalance[10]” reviews how deep learning models handle datasets where one class is underrepresented. It reviews 15 studies, mostly focused on images, and groups the solutions into three types: changing the data, changing the model and combining both. The authors find that deep learning can help, though research is still needed. This study can help us know what to expect when handling a more complex dataset.

3 DATASET DESCRIPTION

The data set we chose is the Spotify Song Attributes Dataset [11]. It comprises over 10,000 songs and includes 523 unique genre labels. For the purposes of this research, the analysis focuses specifically on the top 15 and 151 most frequent genres, which consist of more than 3,000 and 7,000 entries, respectively.

In this study, we focus on musical attributes such as tempo, valence, danceability, etc. They consist of values that are considered relevant for genre classification, while features like id, analysis_url, trackName, etc, are ignored as they do not contribute meaningful information for the classification task.

The most frequently occurring genres in the dataset are alt z and pop, both of them consisting of over 600 entries. The next most common genre, filmi, has slightly more than 400 entries. In contrast, the least represented genres have as few as 10 entries. This unevenness causes a high data imbalance, which can further pose challenges for some classification models.

4 METHOD

In this section we are going to look at the different classification techniques we used throughout the project. We’ll go over the logic behind each method, what makes them suitable for our dataset, and how they contributed to the final performance. Understanding these steps helps make sense of the outcomes we will observe.

4.1 t-SNE

T-SNE is a dimensionality reduction technique that is highly effective for visualizing high-dimensional data in a 2D space. The core idea behind it is that it maps the data based on similarities between data points in the high-dimensional space in order to construct a corresponding distribution in the low-dimensional space. This ensures that the points that are more similar remain closer to each other while those that differ are further apart.

4.2 K-Nearest Neighbors

KNN is a simple and intuitive supervised machine learning algorithm used for solving both classification and regression problems. KNN works by comparing new data points with previously seen examples. It looks at k (a parameter that determines the number of neighbors), then, the algorithm proceeds to compute the distance between the new, unlabeled instance and all instances in the training set. After calculating these distances, the algorithm selects the k instances with the smallest distances. For classification it looks at the most common class among these neighbors, and chooses it as the predicted class.

KNN is often called a lazy learner because it does not learn immediately from the whole training set; instead it stores the dataset and only performs computations at the time of classification.

4.3 Random Forest

Random forest is a widely used algorithm in the field of machine learning for classification tasks. For completing the tasks, it operates by constructing multiple decision trees to make predictions. Each tree is trained on a random subset of the data using random subsets of the features, and each tree makes predictions independently. Finally, all of the predictions from the individual decision trees are used to make a final prediction on the data. This is done through a process called “majority vote”.

Random Forest is particularly effective for genre classification due to its ability to handle high-dimensional data, complex nonlinear patterns and missing values. The algorithm provides a deeper insight into feature importance, helping identify which features that are most influential in distinguishing between genres. It also scales well with large datasets, making it well-suited for our study.

4.4 Support Vector Machine

The support vector machine is a supervised machine learning algorithm used for classification tasks. The core idea of the SVM is to find the hyperplane that best separates data points of different classes. It is chosen in such a way that maximizes the margin between them. This margin represents the distance from the hyperplane to the closest data point on both sides.

SVM employs a technique called “kernels”, which helps map the data in a higher-dimensional space, where a linear separation becomes possible, thereby being able to find a decision boundary even for non-linear data. SVM is particularly well-suited for genre classification due its emphasis on finding the best boundary between classes, which helps separate genres that may have overlapping features.

4.5 Logistic Regression

Logistic regression is a supervised machine learning algorithm used mainly for classification problems. Although its name suggests regression, the goal is actually to predict the probability that a data point belongs to a given class or not, especially in the case where there are only two possible outcomes (yes/no, 0/1 true/false, class A/ class B).

The classifier works by combining input features into a single score using a linear formula. It then applies the sigmoid function that converts the result into a probability between 0 and 1. If the value is above the chosen threshold (typically 0.5) then the algorithm classifies it into class 1, otherwise into class 0.

4.6 Perceptron

The perceptron is one of the simplest artificial neural network models. It works by mapping input features to an output that typically represents either class 0 or class 1. It is composed of one layer of input nodes that are fully connected to an output node.

In a perceptron, a weight is associated with each input node to represent its significance in predicting the output. The perceptron calculates the output by taking the weighted sum of the inputs

and applying an activation function to decide whether it should activate.

A single-layered, while powerful in its simplicity, it struggles with nonlinear problems. Although it is not the best-performing in our study, it still remains a useful foundation, a building block for the more advanced version called the multi-layered perceptron.

4.7 Multi-Layer Perceptron

MLP is an artificial neural network often used for classification tasks. It is an extension of the classical perceptron, adding more hidden layers between the input and the output layers, which enables the network to efficiently work with complex and nonlinear data.

In an MLP, each layer consists of neurons and each neuron computes a weighted sum of its inputs and then applies a nonlinear activation function. The output of one layer becomes the input of another, allowing the network to learn increasingly abstract representations of the data.

Through a process called backpropagation, which is done during training, the MLP minimizes the loss function by adjusting the network's weights and biases. MLPs are versatile and particularly effective for high-dimensional tasks but they are also computationally expensive[12], slow to train and prone to overfitting[13].

The MLP is a highly effective model, due to its ability to handle high-dimensional data which is particularly important when genres share overlapping characteristics.

5 METRICS

5.1 Balanced Accuracy Score

The balanced accuracy score is a performance metric used to evaluate the effectiveness of a classification model, especially when the dataset is imbalanced, meaning some classes have significantly more examples than others. Balanced accuracy provides a more equitable evaluation by accounting for the performance of each class individually.

The balanced accuracy formula is as follows:

$$\text{Balanced Accuracy} = \frac{1}{2} (\text{Sensitivity} + \text{Specificity})$$

Where:

- Sensitivity = $\frac{TP}{TP+FN}$
- Specificity = $\frac{TN}{TN+FP}$

5.2 Receiver Operating Characteristic - Area Under the Curve

The ROC-AUC curve is a tool widely used for evaluating the performance of binary and multi-class classifiers. It plots the ROC curve which represents the true positive rate against the false positive rate showing how well a model distinguishes between classes. The values range from 0 to 1 where: close to 1.0 represents a perfect classifier, 0.5 represents random guessing and close to 0 indicates a performance worse than random. The AUC gives an overall idea of how well the model is distinguishing positives and negatives.

Typically, increasing the number of classes or the volume and diversity of data leads to a more challenging classification task. This effect is seen in the decline of classification metrics, such as balanced accuracy. However, this effect is not as drastic for the AUC

score which tends to decrease more gradually. This phenomenon can be attributed to fundamental differences in what AUC measures and how it is computed.

AUC score measures a model's ability to rank positive instances higher than negative ones. It does not just check if the prediction is right or wrong at a fixed threshold. Instead, it looks at whether the model gives higher scores to the correct class than to the incorrect ones. Hence, even if the model makes some classification errors, it can still achieve a high AUC provided that it consistently ranks the correct class higher than the incorrect ones.

5.3 Recall Score

Recall is a performance metric used to show the proportion of true positives among all actual positives. Recall is particularly important in contexts where false negatives are more harmful or costly than false positives. A high recall score means that the model is very effective at detecting most of the positive cases, which is particularly important in applications where missing a positive instance is costly.

The equation for the Recall score is as follows:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

5.4 Precision Score

Precision is the proportion of true positive predictions among all predicted positives. Precision is particularly important in contexts where false positives are more harmful or costly than false negatives.

The equation for the Precision score is as follows:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

5.5 F1 Score

The F1 score is an evaluation metric that measures a model's performance using the harmonic mean of the precision and recall scores of a model. The harmonic mean is preferred over the arithmetic one because it balances both metrics equally. In a multi-class classification problem, we calculate the F1 score for each class individually rather than calculating a single F1 score for the entire model. This is called the one-vs-rest (OvR) strategy.

The F1 Score is a comprehensive evaluation metric, since it maximizes both the precision and recall scores simultaneously.

The equation for the F1 score is as follows:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

6 RESULTS

In order to analyze the data, we apply the train-test split method to partition the dataset into two distinct subsets. Specifically, 80% of the data is allocated for training purposes, allowing the models to learn underlying patterns, while the remaining 20% is reserved for testing.

This portion of the data is used to assess the model's ability to perform on previously unseen instances, providing a more reliable measure of its effectiveness in real-world scenarios.

According to t-SNE, each point in the plot corresponds to a single song instance from the dataset projected in a two-dimensional space, based on their corresponding features. To facilitate interpretability,

points are colored using distinct colors from the Seaborn ¹palette, using unique colors for each corresponding genre. With the help of this visualization, we can observe clusters of similarly colored points, meaning they share similar audio characteristics, whereas overlapping points that are very close to one another suggest higher inter-genre similarity.

In our study, when using the KNN classifier, we examine the values of k from 3 to 30, to determine which will give the best accuracy for our data. As we can observe, k value gets smaller as the number of classes gets bigger. This may occur because, as the number of classes increases so the distribution of data points changes, and genres with fewer instances will not be classified as accurately. Therefore, the local neighborhood around any given data point is more likely to contain instances from multiple, potentially unrelated, classes. Class imbalance also becomes a problem, as some classes become underrepresented. A smaller k helps with this issue by focusing the decision on the most immediate and potentially more relevant neighbors.

Logistic regression is not the best algorithm for classifying music genres, due to its simplicity and it shows poor results.

The perceptron may also not be the most suitable choice for this study, as it clearly struggles to handle high-dimensional data and complex decision boundaries. It struggles to capture the non-linear relationships often present in real-world datasets such as music genre classification. As seen in the results, its performance tends to lag behind more advanced models that are specifically designed to handle such complexity.

Random forest showed decent results, struggling with the more imbalanced dataset, but it was easily overshadowed by the MLP and SVM. Both classifiers showed top results, close in score. They also managed to keep a remarkable performance when classifying the bigger dataset, showing that they work well with class imbalance.

6.1 Balance Accuracy Results

In the table we can observe the change between the balanced accuracy of each classifier when we increase the number of genres. The most notable trend is the decrease in score that comes with the increase in classes. In multiclass settings, especially with a fixed dataset size, the number of training examples per class decreases as the total number of classes increases leading to a model without the ability to learn class-specific patterns effectively, resulting in lower per-class recall, particularly for rare classes.

As expected, classifiers like KNN and Perceptron significantly decline in performance due to their way of classifying the data. KNN might be influenced by dominant classes, reducing recall for underrepresented ones and thus lowering balanced accuracy. Perceptron will struggle with problems with many classes because it can only find a separating linear hyperplane between classes, making the evaluation a poor one.

Classifiers such as SVM and MLP show a high balanced accuracy and seem to remain the most consistent when increasing the data. Instead of relying on all data points (as KNN does), SVM learns the decision boundary from the most informative examples (support vectors), making it more robust to noise and less biased by dominant classes. SVM handles non-linear decision boundaries through the use of kernel functions, so it works consistently even when feature distributions are highly overlapping. MLPs are capable of automatically learning hierarchical feature representations from the data, which makes them particularly well-suited for tasks involving overlapping or ambiguous genre boundaries.

6.2 ROC AUC Results

Some of the most notable curves are the one made by perceptron, which shows the lowest value, and the one made by SVM with the highest value. Perceptron exhibits poor performance when evaluated. As mentioned before, it relies on the assumption that the dataset is linearly separable and works poorly with high-dimensional data, therefore, its low ROC AUC score is an expected outcome. Unsurprisingly SVM shows a great performance and high effectiveness when classifying high-dimensional data.

Also noteworthy is the performance of the random forest classifier, which is the second best one, right after the SVM. It works this well because it combines the predictions of many decision trees, leading to more stable and accurate probability estimates.

6.3 Recall Results

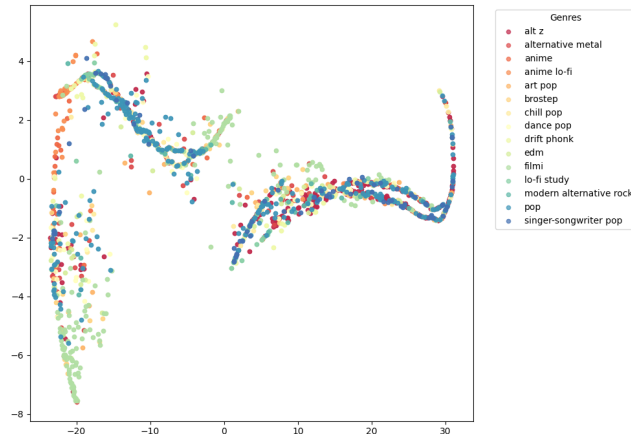
For our study, recall score measures how well the model can identify all songs that truly belong to a specific genre.

KNN, perceptron and logistic regression show surprisingly low results. This is primarily due to some genres having significantly fewer examples than others resulting in classifiers that make biased predictions toward more frequent classes, leading to missed detections of minority classes.

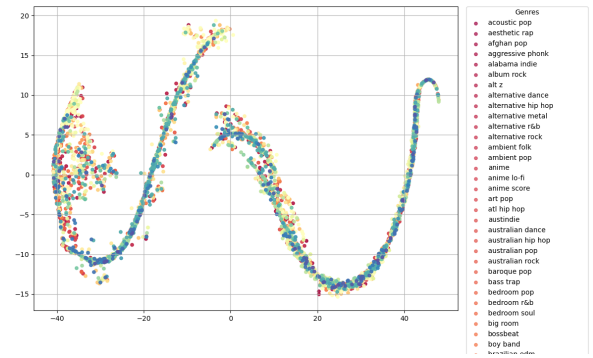
Following the trend, the MLP and SVM are the best performing. Both models are successful in detecting true positives and minimizing false negatives due to their capacity to model complex, non-linear relationships.

¹https://seaborn.pydata.org/tutorial/color_palettes.html

Music Genre Classification

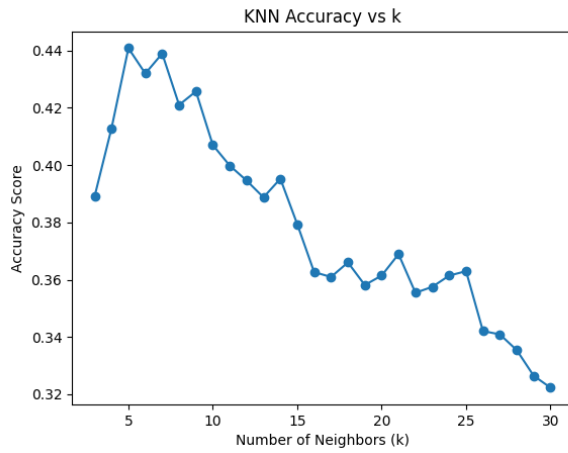


(a) t-SNE Map for 15 Genres

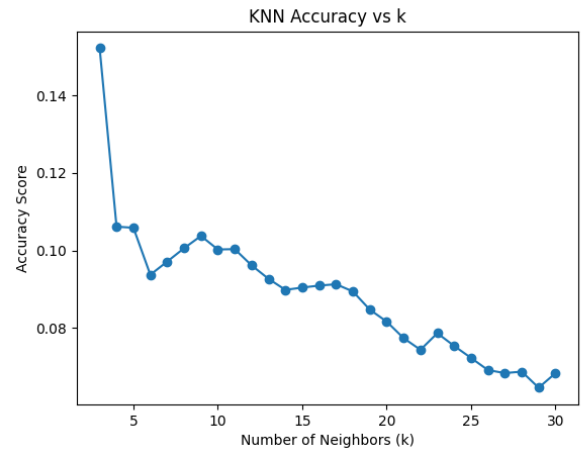


(b) t-SNE Map for 151 Genres

Figure 1: t-SNE Map



(a) Best k for 15 Genres



(b) Best k for 151 Genres

Figure 2: Comparison of best k values.

Metric	RF	KNN	SVM	Perc	MLP	LogR						
							15 Genres					
Balanced Accuracy	0.6507	0.4408	0.8930	0.2760	0.8809	0.4222	0.5158	0.1522	0.8605	0.0484	0.8095	0.1151
Precision	0.8675	0.4794	0.8998	0.3001	0.9032	0.4722	0.8415	0.1416	0.8604	0.0645	0.8175	0.1424
Recall	0.6507	0.4408	0.8930	0.2760	0.8809	0.4222	0.5130	0.1522	0.8605	0.0484	0.8095	0.1151
F1 Score	0.6935	0.4498	0.8936	0.2726	0.8869	0.4231	0.5158	0.1522	0.8605	0.0484	0.8095	0.1151
ROC AUC	0.9790	0.8929	0.9823	0.7645	0.9778	0.8930	0.9737	0.9247	0.9702	0.7598	0.9628	0.8841

Table 1: Comparison of classifier performance on 15 genres and 151 genres. Red values highlight the best score for each metric.

6.4 Precision Results

In the context of our study, precision reflects the model's ability to correctly assign songs to a specific genre without misclassifying songs from other genres as belonging to that genre.

The precision scores across classifiers are generally similar to their corresponding recall scores. Since both metrics assess aspects of the model's ability to make correct positive predictions, it is expected that their values would be closely aligned.

6.5 F1 Results

For the F1 score, simpler classifiers such as logistic regression or k-nearest neighbors tend to show lower F1 scores in imbalanced settings. Due to their ability to handle non linear data, models such as MLP, SVM or random forest tend to maintain a higher F1 score. Their superior performance can be attributed to their flexibility, higher representational capacity, and ability to adapt to imbalanced and non-linear data distributions, which are typical characteristics of real-world genre classification problems.

7 LIBRARIES

To conduct this study, we made use of several widely adopted Python libraries for data analysis, visualization, and machine learning. Specifically, we utilized Pandas² for data manipulation, NumPy³ for numerical operations, and Matplotlib⁴ and Seaborn⁵ for data visualization and plotting. For implementing and evaluating our classification models, we employed Scikit-learn⁶ (sklearn), which provides a comprehensive suite of tools for machine learning tasks, including model training, evaluation, and performance metrics.

8 CONCLUSION

This study explored the task of music genre classification using machine learning techniques applied to the dataset of Spotify Song Attributes. By making use of a range of audio features, we evaluated the effectiveness of several classification models in predicting musical genres across a diverse and imbalanced label set.

Our results demonstrate that complex models such as multi-layer perceptron and support vector machine significantly outperform simpler algorithms in terms of evaluation metrics. Conversely, traditional classifiers such as K-nearest neighbors and perceptron struggled with imbalanced data, resulting in lower predictive performance.

REFERENCES

- [1] Markus Schedl, Emilia Gómez, Julián Urbano. *Music Information Retrieval: Recent Developments and Applications*
<https://julian-urbano.info/files/publications/059-music-information-retrieval-recent-developments-applications.pdf>
- [2] Catherine Guastavino, David Weigl *User studies in the Music Information Retrieval Literature*.
https://www.researchgate.net/publication/220722985_User_studies_in_the_Music_Information_Retrieval_Literature
- [3] Sally Jo Cunningham *Sociology and Music Recommendation Systems*
https://www.academia.edu/26105595/Sociology_and_Music_Recommendation_Systems?
- [4] Anja Volk, Tinka Veldhuis, Katrien Foubert, Jos de Backer *Towards Computational Music Analysis for Music Therapy*
<https://dspace.library.uu.nl/bitstream/handle/1874/436540/000028.pdf?>
- [5] Martin, Rob. *Hit or miss? Predicting Spotify track success from audio feature*.
<https://medium.com/inst414-data-science-tech/hit-or-miss-predicting-spotify-track-success-from-audio-features-9bde821c45dd>
- [6] Shaurya. *EDA - Artists and Genre*.
<https://www.kaggle.com/code/datawithshaurya/eda-artists-and-genres/notebook>
- [7] metarex21. *Spotify Song Attributes Analysis*
<https://github.com/metarex21/SpotXtract/blob/main/spotify-song-attributes-analysis.ipynb>

- [8] Mansisu. *Spotify Tracks Dataset*
<https://medium.com/%40mansisu08/spotify-tracks-dataset-abfd02cdc49>
- [9] Giacomo Rusconi. *Predicting the Popularity of Spotify Songs Using a Neural Network*
https://arno.uvt.nl/show.cgi?fid=171864&utm_
- [10] Mansisu. *Survey on deep learning with class imbalance*
<https://journalofbigdata.springeropen.com/articles/10.1186/s40537-019-0192-5?>
- [11] Byomokesh Senapati. *Spotify Song Attributes*,
<https://www.kaggle.com/datasets/byomokeshsenapati/spotify-song-attributes>
- [12] *Multi-Layer Perceptron Learning in Tensorflow*
<https://www.geeksforgeeks.org/deep-learning/multi-layer-perceptron-learning-in-tensorflow/?>
- [13] Steve Lawrence, C. Lee Giles *Overfitting and neural networks: Conjugate gradient and backpropagation*.
https://www.researchgate.net/publication/3857463_

APPENDIX

²<https://www.geeksforgeeks.org/pandas/introduction-to-pandas-in-python/>

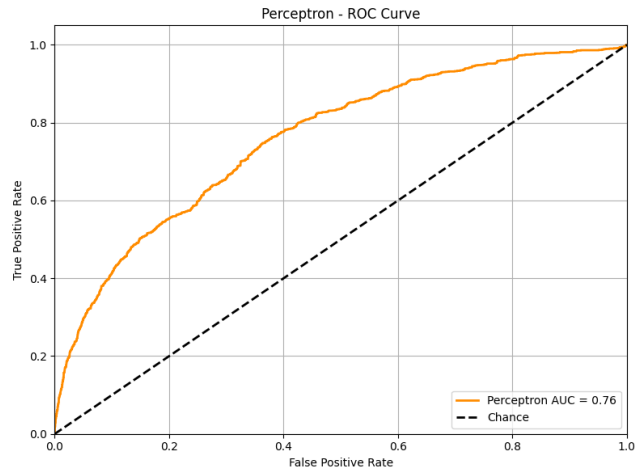
³<https://www.geeksforgeeks.org/python/introduction-to-numpy/>

⁴<https://www.geeksforgeeks.org/python/python-introduction-matplotlib/>

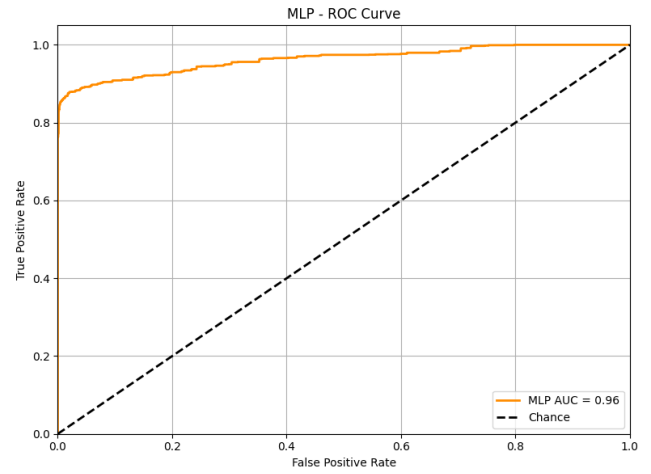
⁵<https://www.geeksforgeeks.org/data-science/seaborn-color-palette/>

⁶<https://scikit-learn.org/stable/>

Music Genre Classification

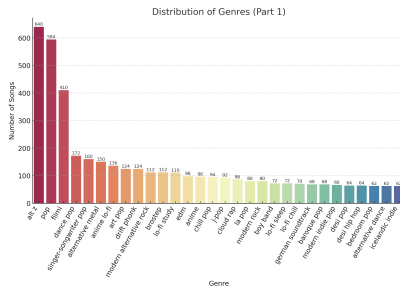


(a) Perceptron ROC Curve for 151 Genres

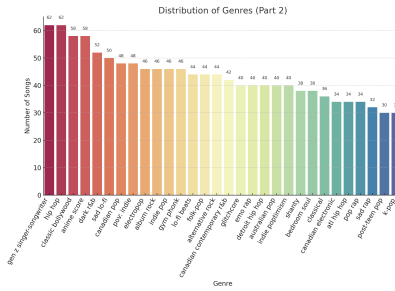


(b) MLP ROC Curve for 151 Genres

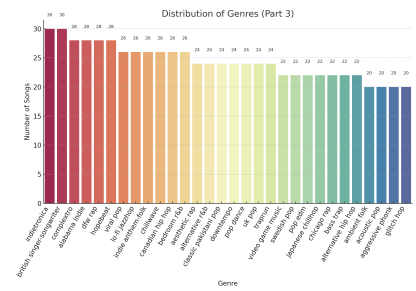
Figure 3: ROC Curves for Perceptron and MLP classifiers on the 151-genre dataset.



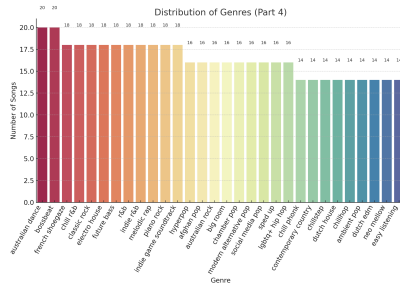
(a) Genre Distribution (Part 1)



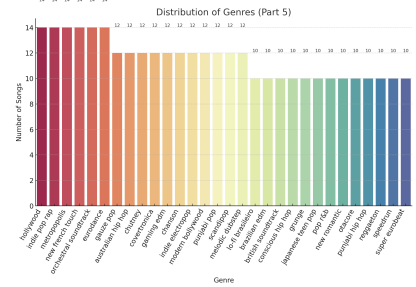
(b) Genre Distribution (Part 2)



(c) Genre Distribution (Part 3)



(d) Genre Distribution (Part 4)



(e) Genre Distribution (Part 5)

Figure 4: Genre distribution visualized in 5 parts due to the high number of classes in the dataset.